

Answer to the Question No-1

we have the display resolution 1920×1080

$\therefore$ The number of pixels $1920 \times 1080 = 20,73,600$ pixels

Answer to the Question No-2

we have An image resolution 1024×768 .

$\therefore$ means $= 786,432$ pixels

here..

1 pixel $= 24$ bits or $24/8 = 3$ bytes

$\therefore$ image size $786,432 \times 3$ bytes
 $= 23,59,296$ bytes

Answer to the Question no-3

Aspect ratio of resolution 1280 x 720 is

$$\begin{aligned} & 1280 : 720 \\ & = 1280/80 : 720/80 \\ & = 16 : 9 \end{aligned}$$

Answer to the Question no-4

Given points $P_1 (2,3)$ and $P_2 (8,9)$

$$\begin{aligned} \text{slope } m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{9 - 3}{8 - 2} = \frac{6}{6} = 1 \end{aligned}$$

straight line equation: $y - y_1 = m(x - x_1)$

$$\Rightarrow y - 3 = 1(x - 2)$$

$$\Rightarrow y = x + 1$$

Answer to the Question no-5

Given pixels (1,1) and (5,9)

Starting pixel (1,1)

Ending pixel (5,9)

$$\therefore \text{slope } m = \frac{9-1}{5-1} = \frac{8}{4} = 2$$

$$\therefore m > 1$$

$$\therefore \frac{1}{m} = \frac{1}{2} = 0.5$$

y	$x = x + \frac{1}{m}$	Round(x)	(x,y)
1	1	1	(1,1)
2	$1 + 0.5 = 1.5$	2	(2,2)
3	$1.5 + 0.5 = 2$	2	(2,3)
4	$2 + 0.5 = 2.5$	3	(3,4)
5	$2.5 + 0.5 = 3$	3	(3,5)
6	$3 + 0.5 = 3.5$	4	(4,6)
7	$3.5 + 0.5 = 4$	4	(4,7)
8	$4 + 0.5 = 4.5$	5	(5,8)
9	$4.5 + 0.5 = 5$	5	(5,9)

Answer to the Question no 6

starting pixel (1,2)

Ending pixel (7,5)

$$\begin{aligned}\therefore \text{slope } m &= \frac{5-2}{7-1} \\ &= \frac{3}{6} \\ &= 0.5\end{aligned}$$

$$\therefore m < 1$$

$$\text{Here, } \Delta x = x_2 - x_1 = 7 - 1 = 6$$

$$2\Delta x = 2 \times 6 = 12$$

$$\text{and } \Delta y = y_2 - y_1 = 5 - 2 = 3$$

$$2\Delta y = 2 \times 3 = 6$$

$$\begin{aligned}\therefore P_0 &= 2\Delta y - \Delta x \\ &= 6 - 6 \\ &= 0\end{aligned}$$

$$\begin{aligned}P_3 &= P_2 + 2\Delta y - 2\Delta x \\ &= 0 + 6 - 12 \\ &= -6\end{aligned}$$

$$\begin{aligned}\therefore P_1 &= P_0 + 2\Delta y - 2\Delta x \\ &= 0 + 6 - 12 \\ &= -6\end{aligned}$$

$$\begin{aligned}P_4 &= P_3 + 2\Delta y \\ &= -6 + 6 \\ &= 0\end{aligned}$$

$$\begin{aligned}\therefore P_2 &= P_1 + 2\Delta y \\ &= -6 + 6 \\ &= 0\end{aligned}$$

$$\begin{aligned}P_5 &= P_4 + 2\Delta y - 2\Delta x \\ &= 0 + 6 - 12 \\ &= -6\end{aligned}$$

$$P_6 = P_5 + 2\Delta y$$

$$= -6 + 6$$

$$= 0$$

K	P_K	x	y	(x,y)
0	0	1	2	(1,2)
1	-6	2	3	(2,3)
2	0	3	3	(3,3)
3	-6	4	4	(4,4)
4	0	5	4	(5,4)
5	-6	6	5	(6,5)
6	0	7	5	(7,5)

Answer to the Q no-7

Given that Radius, $r=7$

Initial point, $x=0$, $y=7$

$$P_0 = 1 - r$$

$$= 1 - 7$$

$$= -6$$

$$\therefore P_K < 0$$

$$\begin{aligned}
 P_1 &= P_0 + 2x_0 + 3 \\
 &= -6 + 0 + 3 \\
 &= -3
 \end{aligned}$$

$$\begin{aligned}
 P_2 &= P_1 + 2x_1 + 3 \\
 &= -3 + 2 + 3 \\
 &= 2
 \end{aligned}$$

$$\therefore P_k > 0$$

$$\begin{aligned}
 P_3 &= P_2 + 2x_2 - 2y_2 + 5 \\
 &= 2 + 4 - 14 + 5 \\
 &= -3
 \end{aligned}$$

$$\begin{aligned}
 P_4 &= P_3 + 2x_3 + 3 \quad [P_3 < 0] \\
 &= -3 + 6 + 3 \\
 &= 6
 \end{aligned}$$

$$\begin{aligned}
 P_5 &= P_4 + 2x_4 - 2y_4 + 5 \quad [P_4 > 0] \\
 &= 6 + 8 - 12 + 5 \\
 &= 7
 \end{aligned}$$

k	P_k	x	y	x, y
0	-6	0	7	(0, 7)
1	-3	1	7	(1, 7)
2	2	2	7	(2, 6)
3	-3	3	6	(3, 6)
4	6	4	6	(4, 5)
5	7	5	5	(5, 5)

Answer to the Question no 8.

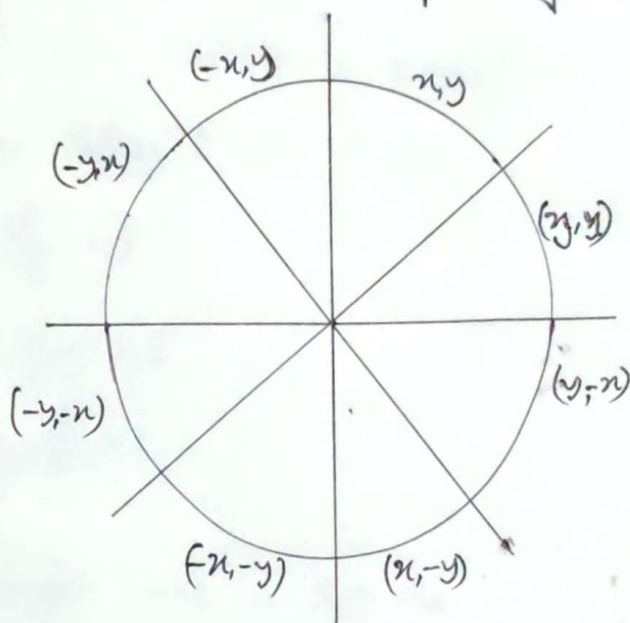
From Question 7

The points in the first octant are:

$(0,7), (1,7), (2,6), (3,6), (4,5), (5,5)$

using 8 way symmetric, The corresponding points are:

1. (x,y)
2. (y,x)
3. $(x,-y)$
4. $(-x,y)$
5. $(-y,x)$
6. $(y,-x)$
7. $(-x,-y)$
8. $(y,-x)$



(x,y)	(y,x)	$(x,-y)$	$(-x,y)$	$(-y,x)$	$(-y,-x)$	$(-x,-y)$	$(y,-x)$
$(0,7)$	$(7,0)$	$(0,-7)$	$(0,7)$	$(-7,0)$	$(-7,0)$	$(0,-7)$	$(7,0)$
$(1,7)$	$(7,1)$	$(1,-7)$	$(-1,7)$	$(-7,1)$	$(-7,-1)$	$(-1,-7)$	$(7,-1)$
$(2,6)$	$(6,2)$	$(2,-6)$	$(-2,6)$	$(-6,2)$	$(-6,-2)$	$(-2,-6)$	$(6,-2)$
$(3,6)$	$(6,3)$	$(3,-6)$	$(-3,6)$	$(-6,3)$	$(-6,-3)$	$(-3,-6)$	$(6,-3)$
$(4,5)$	$(5,4)$	$(4,-5)$	$(-4,5)$	$(-5,4)$	$(-5,-4)$	$(-4,-5)$	$(5,-4)$
$(5,5)$	$(5,5)$	$(5,-5)$	$(-5,5)$	$(-5,5)$	$(-5,-5)$	$(-5,-5)$	$(5,-5)$

Answer to the Question no-20

Given that,

$$\text{Radius on } x\text{-axis } (r_x) = 5$$

$$\text{Radius on } y\text{-axis } (r_y) = 3$$

a) The standard equation of an ellipse centered at $(0,0)$ is:

$$\frac{x^2}{r_x^2} + \frac{y^2}{r_y^2} = 1$$

Substituting the values:

$$\frac{x^2}{r_x^2} + \frac{y^2}{r_y^2} = 1$$

$$\Rightarrow \frac{x^2}{5^2} + \frac{y^2}{3^2} = 1$$

$$= \frac{x^2}{25} + \frac{y^2}{9} = 1$$

$$\begin{aligned} \text{b) length of major axis} &= 2 \times r_x \\ &= 2 \times 5 \\ &= 10 \text{ units} \end{aligned}$$

$$\begin{aligned} \text{length of minor axis} &= 2 \times r_y \\ &= 2 \times 3 \\ &= 6 \text{ units} \end{aligned}$$

c) In the midpoint ellipse, drawing Algorithm, Region 1 (Q1) and region 2 (Q2) begins when the absolute slope of the curve become greater than 1.

The break condition is defined as : $2r_y^x x \geq 2r_x^y y$

Substituting the values : $2 \cdot 3^x x \geq 2 \cdot 5^y y$

$$\Rightarrow 2 \cdot 9x \geq 2 \cdot 25y$$

$$\Rightarrow 18x \geq 50y$$

d) Given that,

Initial point $(0, r_y) = (0, 3)$

Region 1 calculation:

$$r_x = 5, \quad r_x^x = 25$$

$$r_y = 3, \quad r_y^y = 9$$

$$\therefore P1_0 = r_y^y - (r_x^x r_y) + \left(\frac{1}{4} r_x^x\right)$$

$$= 9 - (25 \times 3) + (0.25 \times 25)$$

$$= -59.75$$

$$\therefore P1_1 = P1_0 + 2r_y^y x_1 + r_y^x$$

$$= -59.75 + 2 \times 9 \times 1 + 9$$

$$= -32.75$$

$$P_{12} = P_{11} + 2\tilde{r}_y^x x_2 + \tilde{r}_y^y$$

$$= -32.75 + 2 \times 9 \times 2 + 9$$

$$= 12.25$$

$$P_{13} = P_{12} + 2\tilde{r}_y^x x_3 + \tilde{r}_y^y$$

$$= 12.25 + 2 \times 9 \times 3 + 9$$

$$= -24.25$$

$$P_{14} = P_{13} + 2\tilde{r}_y^x x_4 + \tilde{r}_y^y$$

$$= -24.25 + 2 \times 9 \times 4 + 9$$

$$= 56.25$$

$$P_{15} = P_{14} + 2\tilde{r}_y^x x_5 + 2\tilde{r}_x^y y_5 + \tilde{r}_y^y$$

$$= 56.25 + 2 \times 9 \times 5 - 2 \times 25 \times 1 + 9$$

$$= 105.25$$

K	P_{1K}	x	y	(x, y)	$2\tilde{r}_y^x x$	$2\tilde{r}_x^y y$
0	-59.75	0	3	(0, 3)	0	150
1	-32.75	1	3	(1, 3)	18	150
2	12.25	2	3	(2, 3)	36	150
3	-24.25	3	2	(3, 2)	54	100
4	56.25	4	2	(4, 2)	72	100
5	105.25	5	1	(5, 1)	90	50

since $30 > 50$ ($2r_y^x n > 2r_x^y y$) Region 1 ends here

Region 2:

Initial Decision parameter,

$$\begin{aligned} P2_0 &= r_y^x (n+0.5)^x + r_x^y (y-1)^y - r_n^x n^x - r_y^y y^y \\ &= 9(5+0.5)^x + 25(1-1)^y - 25 \cdot 9 \\ &= 47.25 \end{aligned}$$

since $P2_k > 0$ the next y decreases by 1 and n stay the same, $n=5$
 $\underline{y} = 1-1 = 0$

$$\begin{aligned} P2_1 &= P2_0 - 2r_x^y y_0 + r_n^x \\ &= 47.25 - 2 \times 25 \times 0 + 25 \\ &= 72.25 \end{aligned}$$

K	$P2_k$	n	y	(n, y)	$2r_y^x n$	$2r_x^y y$
0	47.25	5	1	(5, 1)	30	50
1	72.25	5	0	(5, 0)	30	0

we stop here because $\underline{y} = 0$

Answer to the Question no-11

a) Given that

$$x^2 + y^2 = 49$$

comparing with $x^2 + y^2 = r^2$

$$\therefore r^2 = 49$$

$$\therefore \text{radius } r = 7$$

b) Given point $(2, -3)$

substitute $x=2$, $y=-3$ into the equation

$$x^2 + y^2 = 2^2 + (-3)^2$$

$$\Rightarrow 4 + 9 = 13$$

since $13 < 49$ [$r^2 = 49$ from a]

the point $(2, -3)$ is inside the circle.